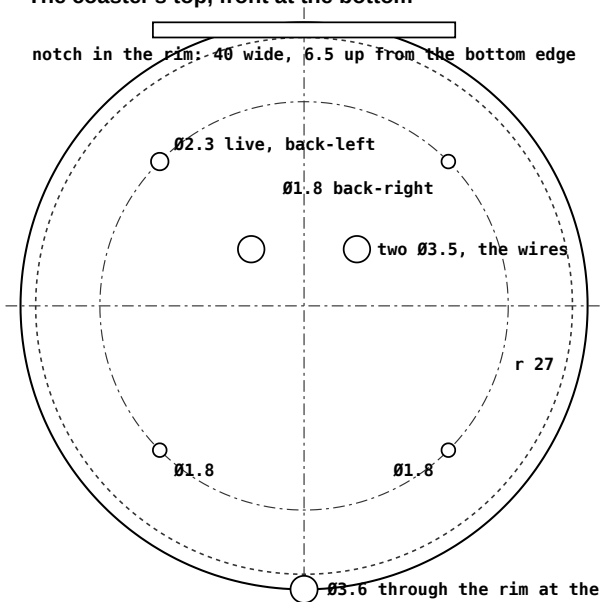
Desk lighthouse: templates at actual size

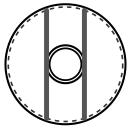
Print landscape on Letter at Actual size: the 50 mm bar measures 50 mm. Prick the hole centres through the paper onto the brass; drill with the cordless drill at its slowest, saw with the jeweler's saw, file to the line.

The coaster's top, front at the bottom

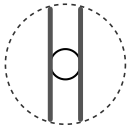


The feet sit on the Ø54 circle at 45°; the wire holes sit between the motor and the tray, 7 either side of the axis and 7.5 behind it. Drill the 1.8, 2.3 and 3.5 straight through the 1.3 mm top; the Ø3.6 for the button goes through the rim at the front. Saw the 40 mm notch from the rim's bottom edge at the back with two cuts, break it out, file.

The spine's hardware: 1 mm rod and tube rings, actual size



the bearing wheel, on the deck
rim: a 3 mm ring of the 5/8 in tube
hub: a 3 mm ring of the 5 mm tube
two spokes 14.3, tangent to the hub



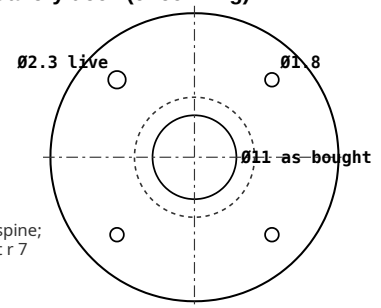
the shutter's floor
two bars 14.6, tangent to the spine,
0.5 mm up inside the drum (dashed)



the magnet arm
6.5 of 1 mm rod on the spine;
the magnet under it at r 7

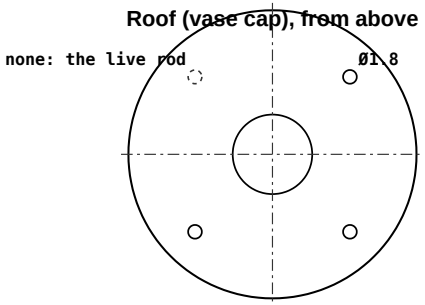
Five pieces of 1 mm rod (64 mm in all) and two rings; file the ring ends square.
Make the wheel on the spine itself: the hub ring on the spine, the spokes soldered to its flanks,
the rim ring to the spoke ends; it centres itself. Slide it down onto the deck and glue the rim.
The bars go on last, with the drum: solder them to the spine's flanks, then their ends to the wall.

Gallery deck (check ring)



The four rods pass at r 14.5, at 45° (front at the bottom).
Dashed: the wheel's rim Ø15.9 over the Ø11 hole;
the live rod's hole, back-left, takes its sleeve.

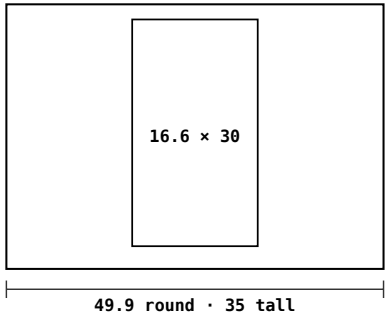
Roof (vase cap), from above



Three Ø1.8 holes at r 14.5, on the dome's slope:
centre-punch, drill from outside; the rods pass
through and are soldered flush. Measure the cap first.

50 mm: check it with a ruler

The shutter, unrolled: 5/8 in brass tube



120° window, 3 up from the bottom end. Wrap the strip round the tube, prick the corners, saw, file.
A 3 mm cut of the same tube is the wheel's rim.

One rod, not to scale



cut 223 · 6 mm foot · deck mark 174.5 from the bend
live rod: sleeves at foot and deck, boot at the tip, 1.4 shorter

Order of work

1. Coaster: four rod holes, two wire holes, the notch at the back, the button's Ø3.6 in the rim at the front. Roof: three Ø1.8 holes in the dome. Shutter: cut 35, saw the window, plus a 3 mm ring from the same tube. Charger: jumper cut.
2. From 1 mm rod: two bars, two spokes, the arm; a 3 mm ring of the 5 mm tube. Make the wheel on the spine as a jig: hub ring, spokes on its flanks, rim ring on their ends. Solder the arm to the spine 6.5 above where the gearbox face sits.
3. Bend the motor's tabs, solder its wires, set it on its bed, shaft on the axis. Slide the spine over the D shaft with a drop of gel glue; sight it true as the motor turns slowly. Glue the magnet under the arm and the reed behind the motor.
4. Rods: sleeve the live one; thread each through the coaster hole, bend the foot under, glue. Slide the deck down to 168, nudge the rods vertical and glue it; slide the wheel down onto the deck, glue its rim over the hole; trim the live rod.
5. Shutter: solder the bars to the spine's flanks 0.5 mm inside the drum's bottom edge, window to the front at the index, then the bar ends to the wall from below. Lantern: post into the nipple, filament on the post, nut, then the cap down over the three rods' tips, soldered flush outside, the live rod booted beneath; the ball; two wires to the back rods.
6. Boards, diode, divider, the button inside the rim, the washers in the hollow, both ports in the notch. The tray behind the motor on a dot of glue, leads down the left hole; the cell in. Flash through the notch; then the meter checks.

Rev D, drawn 2026-11-09 evening for approval. The deck assumes the 1-1/2 in check ring, the roof the 1-1/2 in vase cap, and the coaster's top 1.3 mm from its listed weight. Measure the real parts and rerun the script before cutting.